

RG-D Experiments: Status and Analysis Plans

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CLAS Collaboration (18–21 November 2025)

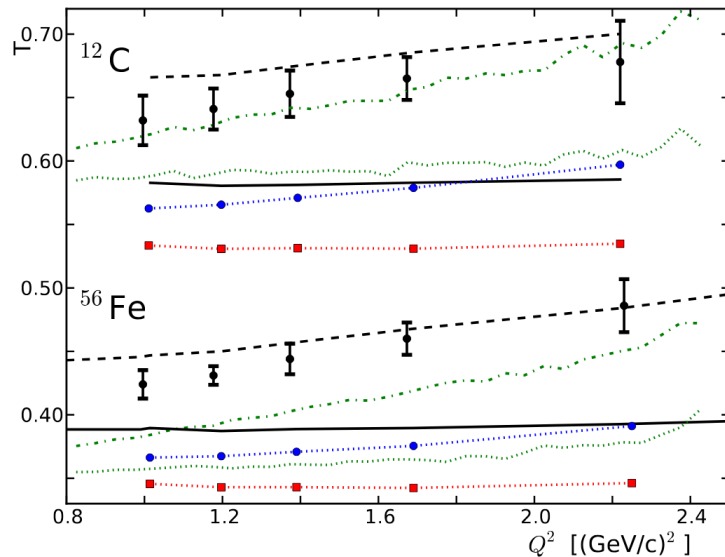


Run Group D

RG-D is composed of two experiments:

Study of Color Transparency (CT) in Exclusive Vector Meson Electro-production off Nuclei (E12-06-106):

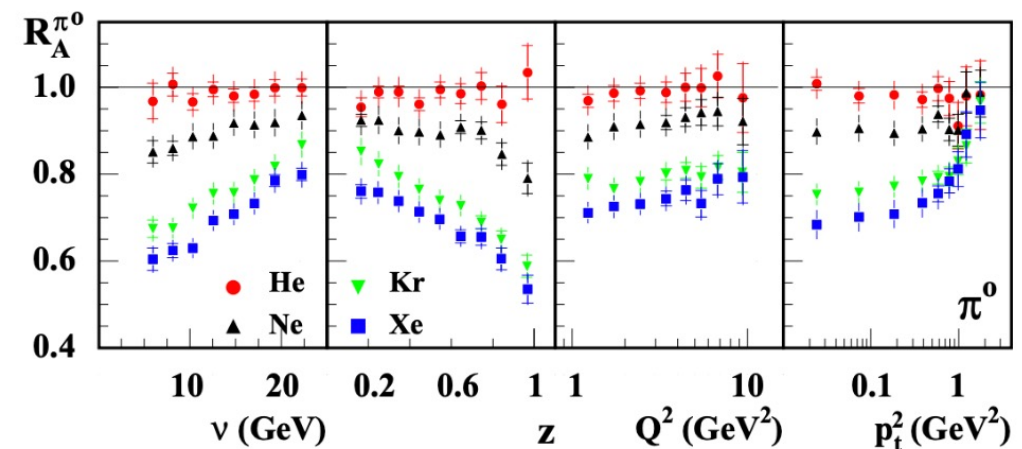
Spokespeoples: W. Armstrong¹, L. El Fassi³, K. Hafidi¹, M. Holtrop⁴, and B. Mustapha¹



W.~Cosyn and J.~Ryckebusch
arXiv:1301.1904

Nuclear Transverse Momentum Distributions (nTMDs) in CLAS12 (E12-06-106A):

Spokespeoples: R. Dupré², L. El Fassi³, Zein-Eddine Meziani¹, and Holly Szumila-Vance⁵



HERMES Collaboration
arXiv:0704.3270

¹: Argonne National Lab, ²: IJCLAB, Orsay, France ³: Mississippi State U., ⁴: University of New-Hampshire, ⁵: Jefferson Lab

RG-D: CT Experiment

- E12-06-106: Color Transparency: suppression of interactions of colorless small size configuration in nuclear medium
- Experimental signature: rise of the nuclear transparency, T_A as a function of Q^2 :
 - T_A : ratio of the cross section per nucleon on a bound nucleon to that on a free nucleon

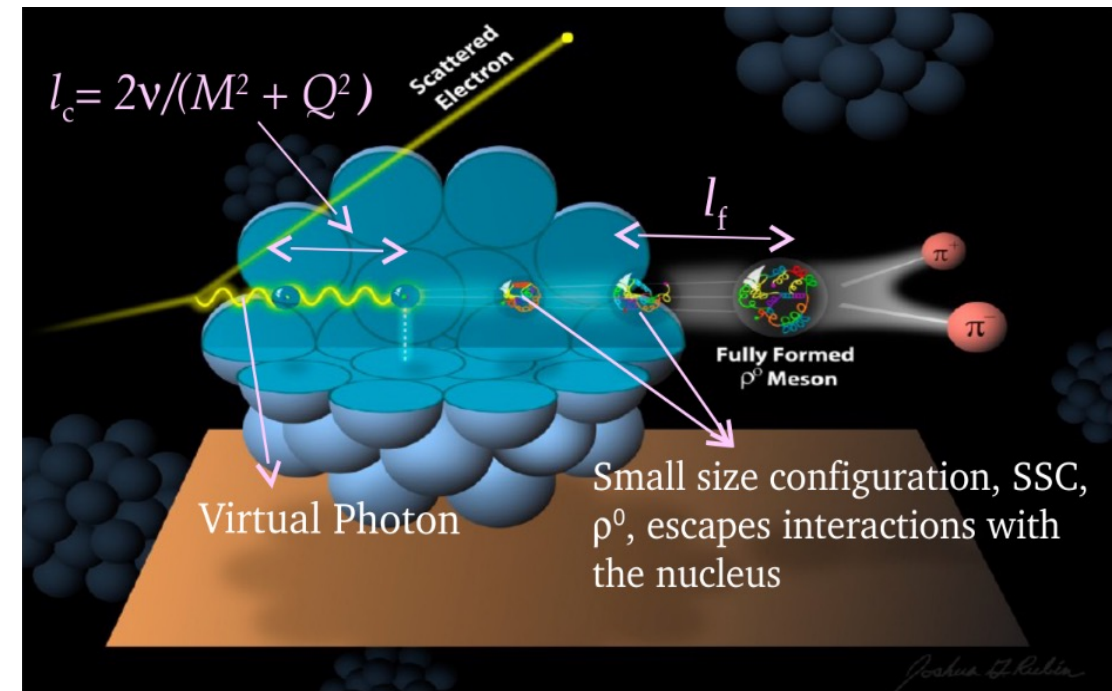
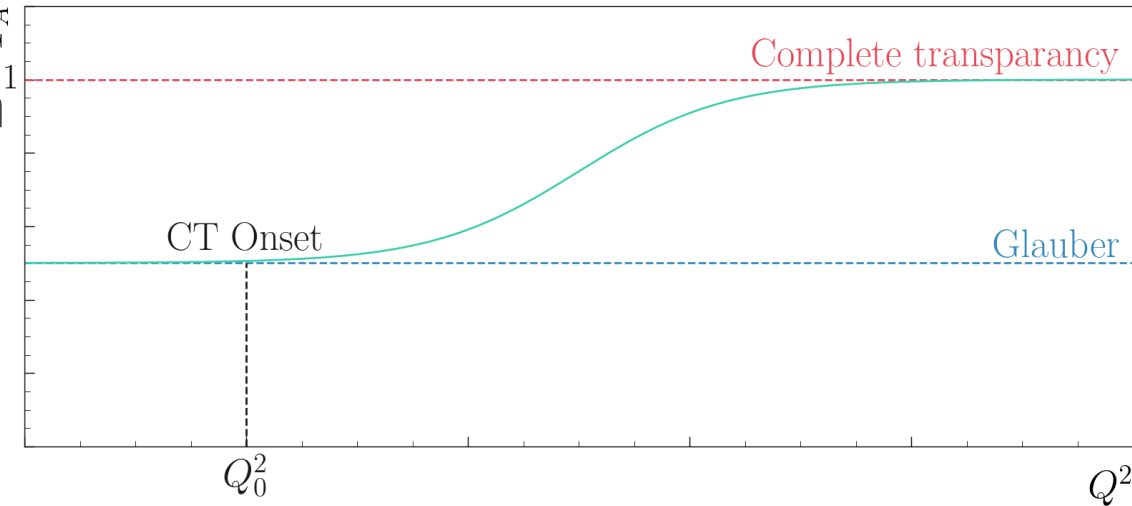
$$T_A = \frac{\sigma_A}{A\sigma_N}$$

σ_A – nuclei cross section
 σ_N – free nucleon cross section

Objective: understanding the dynamical evolution from small-size configurations into ordinary hadrons, and validating the QCD factorization theorem

Coherence length, l_c : the lifetime of the $q\bar{q}$ -pair

Formation time, l_f : the time evolution of SSC to an on-shell ρ^0 meson



RG-D: Nuclear TMDs

- E12-06-106A: nTMDs study uses the same CT running conditions except the beam polarization, and aims to study:

- Nuclear effect on SIDIS asymmetries
- Transport coefficient of the nuclear matter

- For that, measure different observables:

- $\cos \varphi$, $\sin \varphi$, $\cos(2\varphi)$ ¹ moments for different hadrons (π^+ , π^- , and π^0):

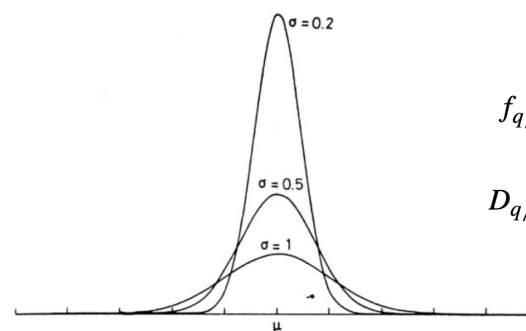
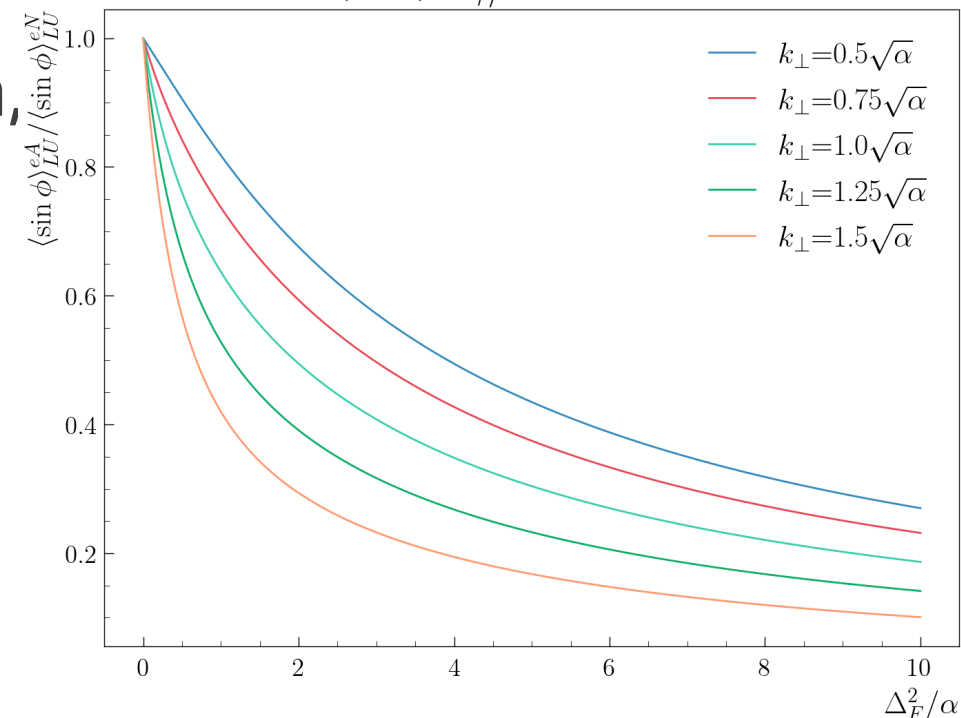
$$\frac{\langle \cos \phi \rangle_{UU}^{eA}}{\langle \cos \phi \rangle_{UU}^{eN}} \approx \frac{\alpha}{\alpha + \Delta_{2F}}, \quad \frac{\langle \sin \phi \rangle_{LU}^{eA}}{\langle \sin \phi \rangle_{LU}^{eN}} \approx \frac{\alpha}{\alpha + \Delta_{2F}}.$$

$$\Delta_{2F} = \int d\xi_N^- \hat{q}_F(\xi_N),$$

$\hat{q}_F(\xi_N)$: the quark transport parameter

¹ No predictions exist for this observable

Song, Y. et al. (2014) Physical Review D, 89 014005.



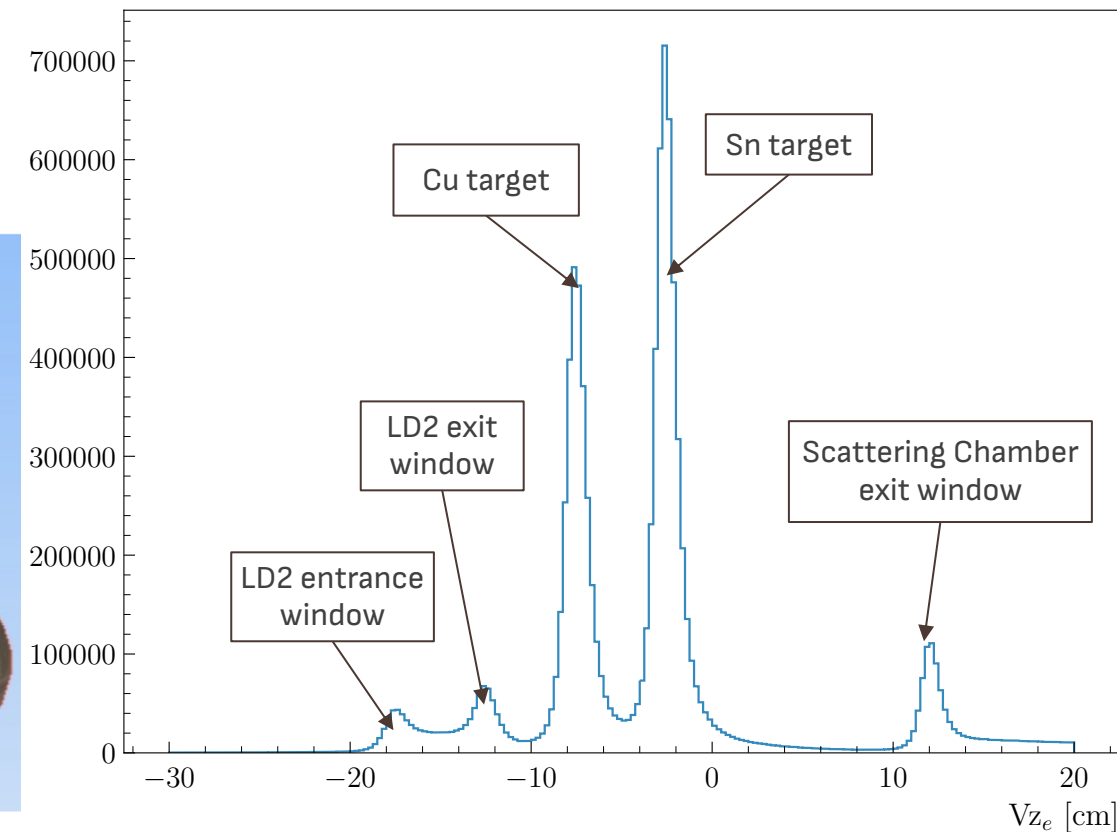
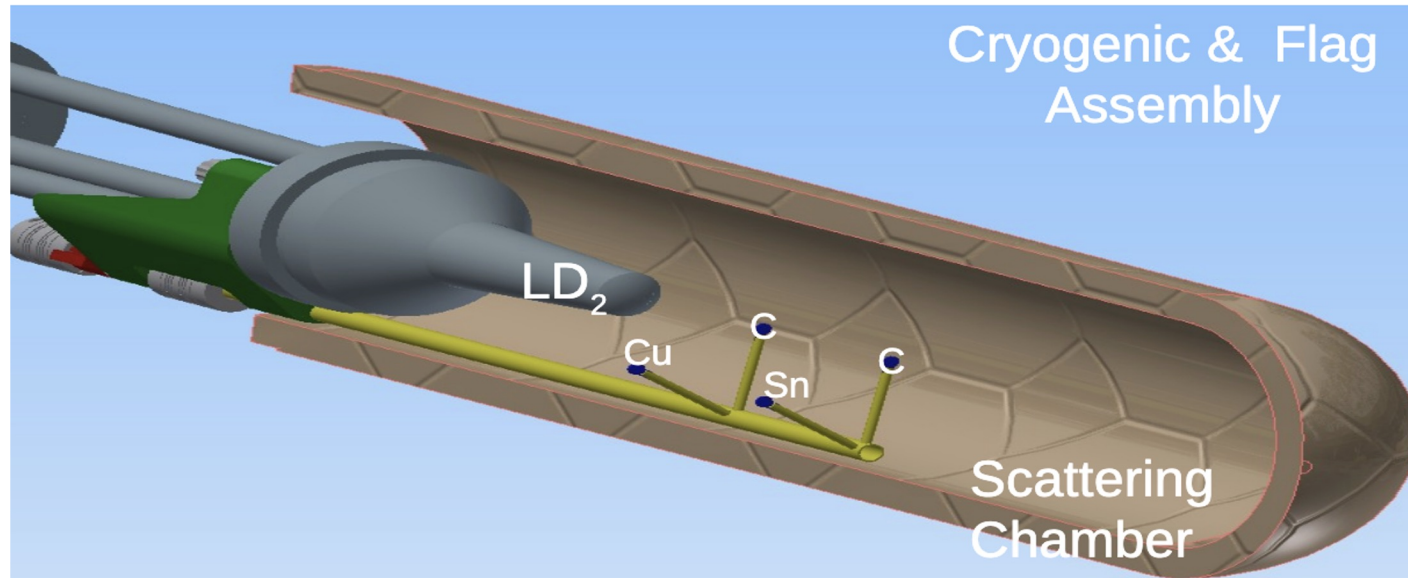
$$f_{q/p}(x, k_T, Q) \simeq \frac{1}{\pi \langle k_T^2 \rangle} \exp\left(-\frac{k_T^2}{\langle k_T^2 \rangle}\right) f_{q/p}(x, Q)$$

$$D_{q/p}(z, p_T, Q) \simeq \frac{1}{\pi \langle p_T^2 \rangle} \exp\left(-\frac{p_T^2}{\langle p_T^2 \rangle}\right) D_{q/p}(z, Q)$$

The heavier the nucleus, the wider the TMD gaussian width (Nuclear Broadening)

RG-D Run Configuration

- RG-D experiments collected data in the fall of 2023 with
 - 10.54 GeV polarized beam energy and $I=5\text{--}175$ nA for luminosity scans and production
 - Standard CLAS12 configuration with FT-OFF and three layers of FMT
 - Three target configurations, LD2, CxC, and CuSn, deployed with InBending and OutBending
 - New cryogenic LD2 and the nuclear-foil flag assembly centered at -5 cm for each configuration
- Main run hiccups:
 - Faraday Cup vacuum contamination
 - Moller cone sagging



RG-D Status: Calibration Summary

- RG-D successfully passed Pass1 review on May 27th
- Pass1 cooking initiated on June 13th using COATJAVA 13.0.0
- The cooking is at 49%:
 - Manly due to the high usage of the farm and recent modification of the job scheduling

Overall Progress

49.3%

87275 of 177186 jobs completed

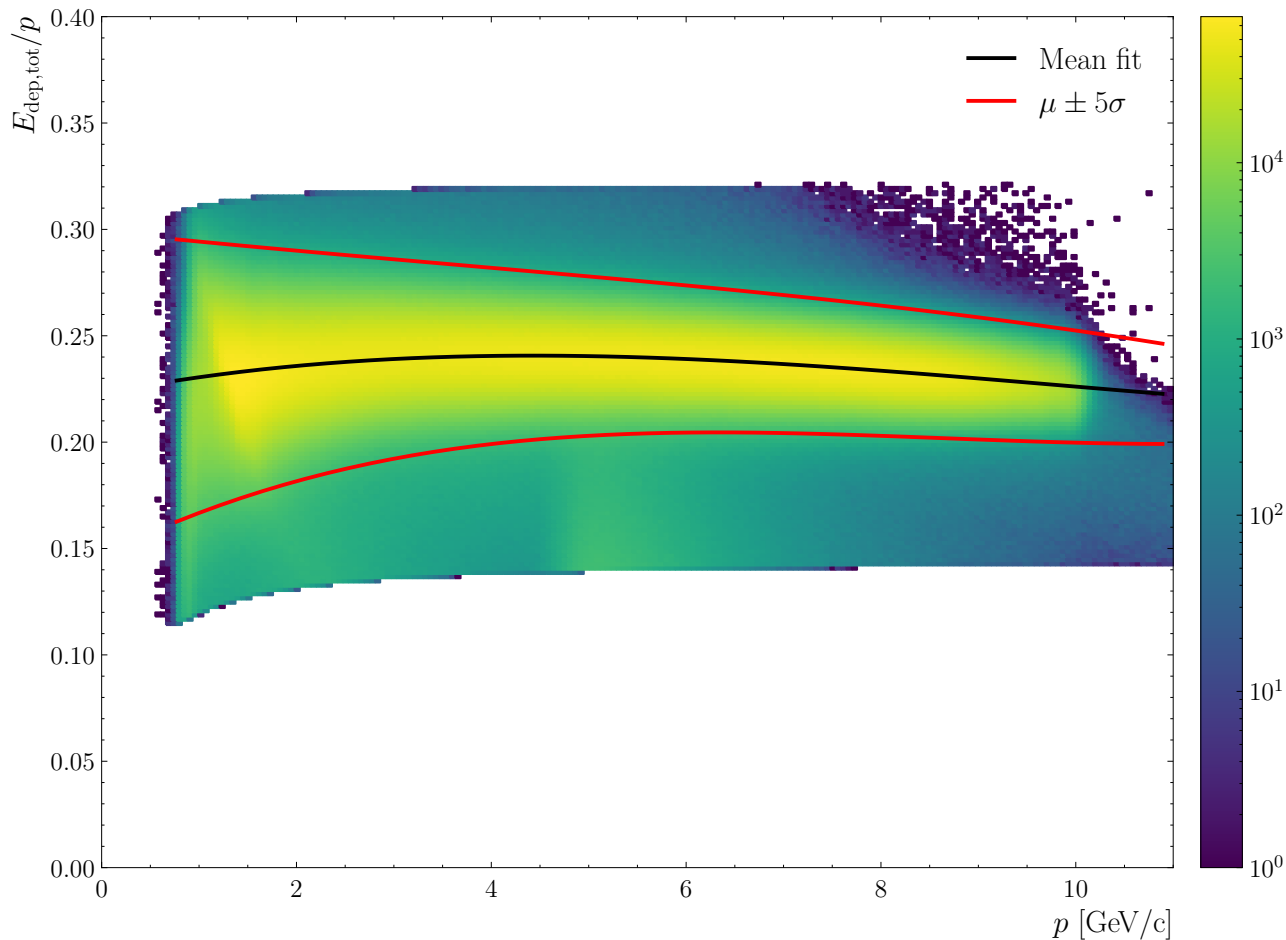
Thanks to CalCom Experts for their support

Electron fiducial cuts

- Since the last meeting, refinement cuts have been developed and finished to improve electron identification
- Raw electron cuts:
 - $\text{pid} = 11, \text{status} < 0$
 - $2000 \leq |\text{status}| < 4000$
 - $|\text{chi2pid}| < 5$
- Refinements:
 - Sampling fraction vs momentum: should be constant for an electron
 - v and w for the calorimeter: defined at lengths that are multiples of 4.5cm
 - DC fiducial cut using $\text{chi2}/\text{ndf}$ vs edge: remove the edges of each region
 - V_z cut: select an electron that comes from the target
 - $p > 0.8 \text{ GeV}/c$: reconstruction below 0.8 GeV can't be trusted

Electron fiducial cuts: Sampling fraction vs p

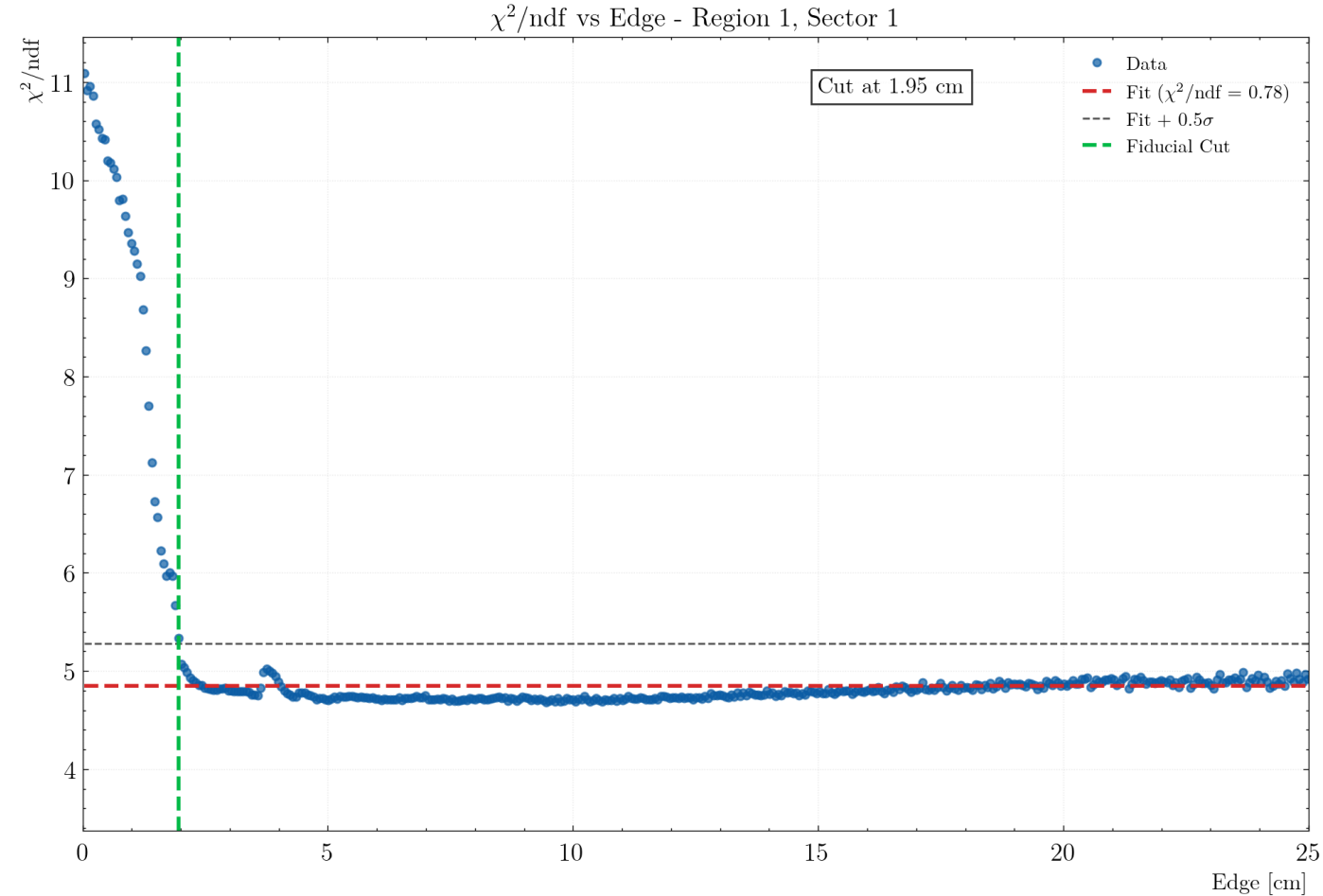
- This cut is done by extracting μ and σ for 50 p slices by fitting with Gaussian.
- Then fitting them with 3rd order polynomial function. This cut is sector, polarity and target dependent



Sector	Parameter	LD2	CuSn	CxC
1	a_μ	0.23888	0.25010	0.23861
	b_μ	0.0031921	-0.0039137	0.0034309
	c_μ	-0.00058438	0.00076421	-0.00062000
	d_μ	1.3664×10^{-5}	-7.2623×10^{-5}	1.5512×10^{-5}
	a_σ	0.025201	0.023303	0.024357
	b_σ	-0.0047077	-0.0027082	-0.0042778
	c_σ	0.00052813	0.00018090	0.00046235
	d_σ	-2.2138×10^{-5}	2.7946×10^{-7}	-1.8943×10^{-5}
2	a_μ	0.23746	0.24910	0.23747
	b_μ	0.0026166	-0.0037407	0.0030234
	c_μ	-0.00024773	0.0015984	-0.00031869
	d_μ	-1.0829×10^{-5}	-1.8937×10^{-4}	-6.7700×10^{-6}
	a_σ	0.024503	0.032076	0.023522
	b_σ	-0.0042768	-0.0088127	-0.0038011
	c_σ	0.00045956	0.0014080	0.00039416
	d_σ	-1.7030×10^{-5}	-5.6450×10^{-5}	-1.4115×10^{-5}

Electron fiducial cuts: DC Fiducial cuts

- DC Fiducial cuts are applied on the variable edge and are define using χ^2/ndf vs edge



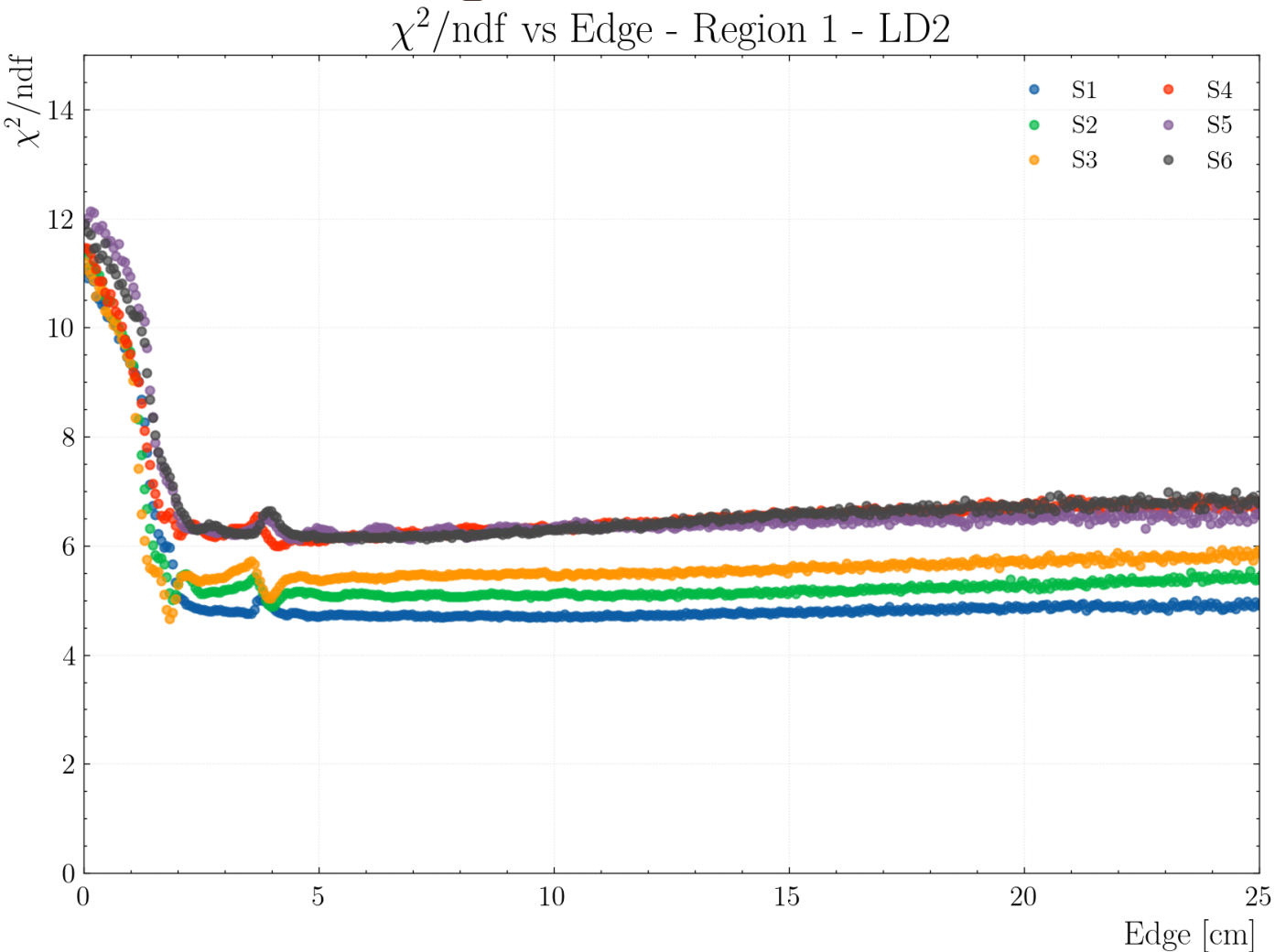
Strategy:

- Fit the part between 13 and 25 with a straight line
- Get the distance between the fit and each data points, compute the std dev
- Find 5 consecutive points above the line + std dev starting from the right

Sector independent cut, but different for each particle and target

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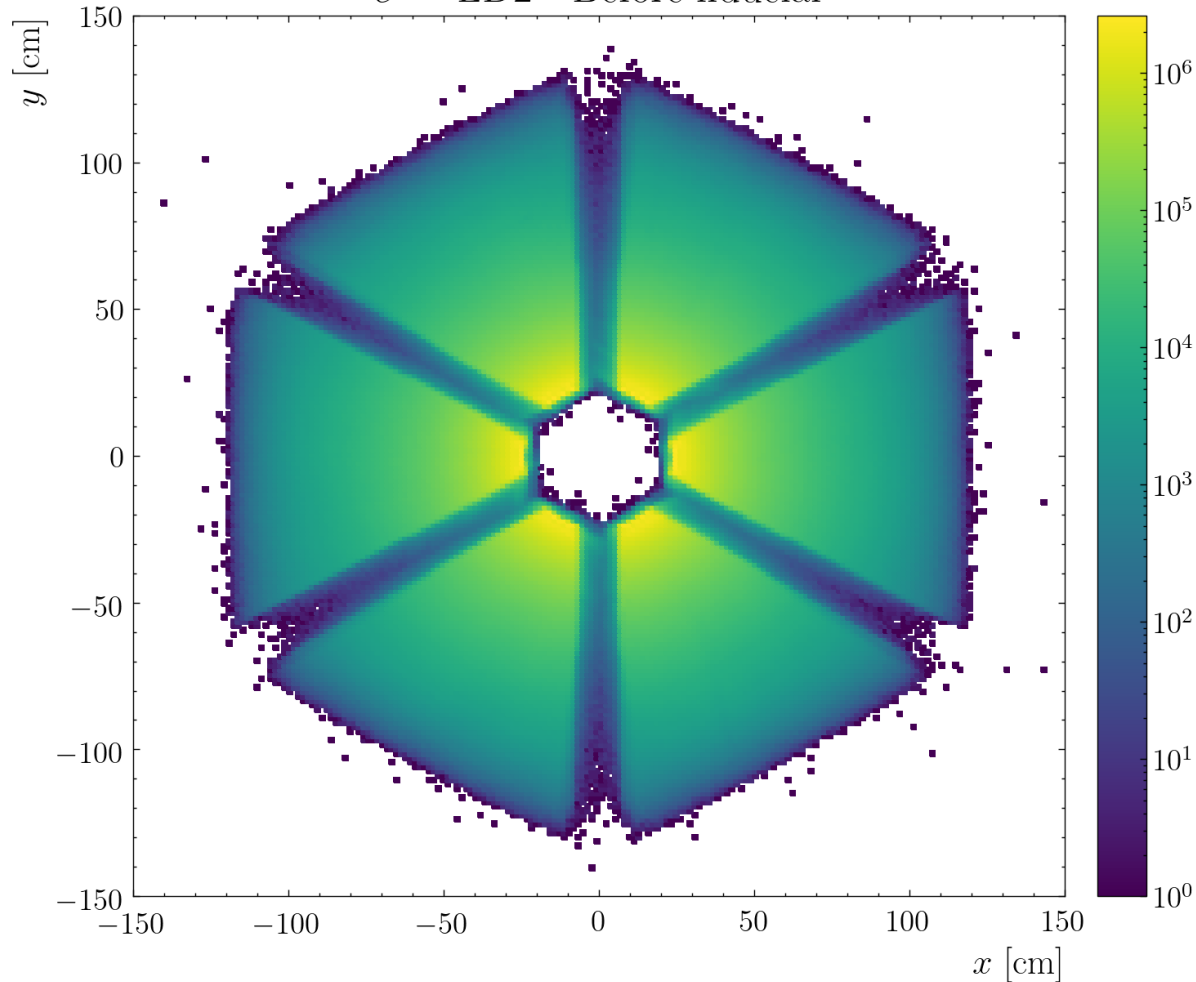
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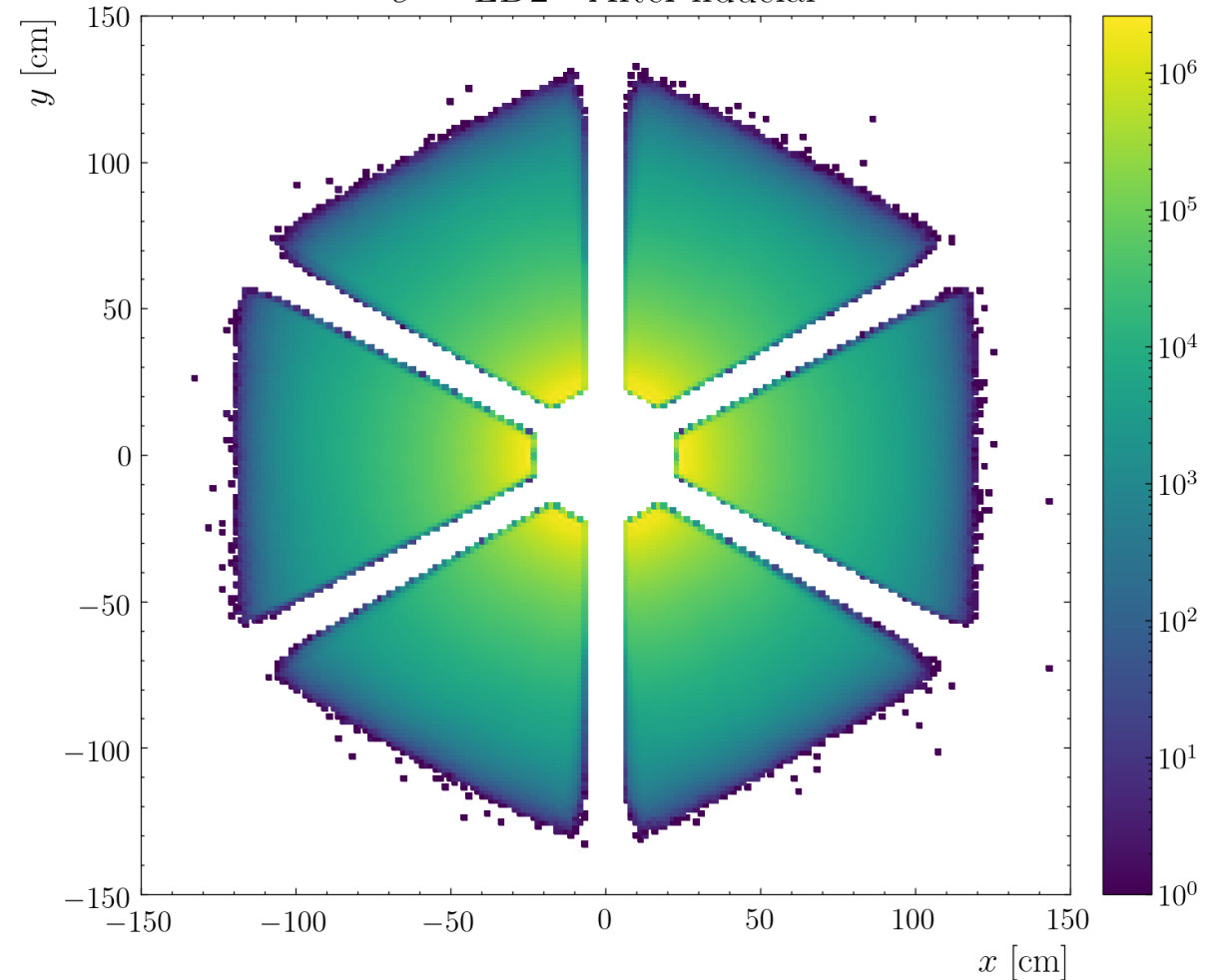
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e^- - LD2 - Before fiducial

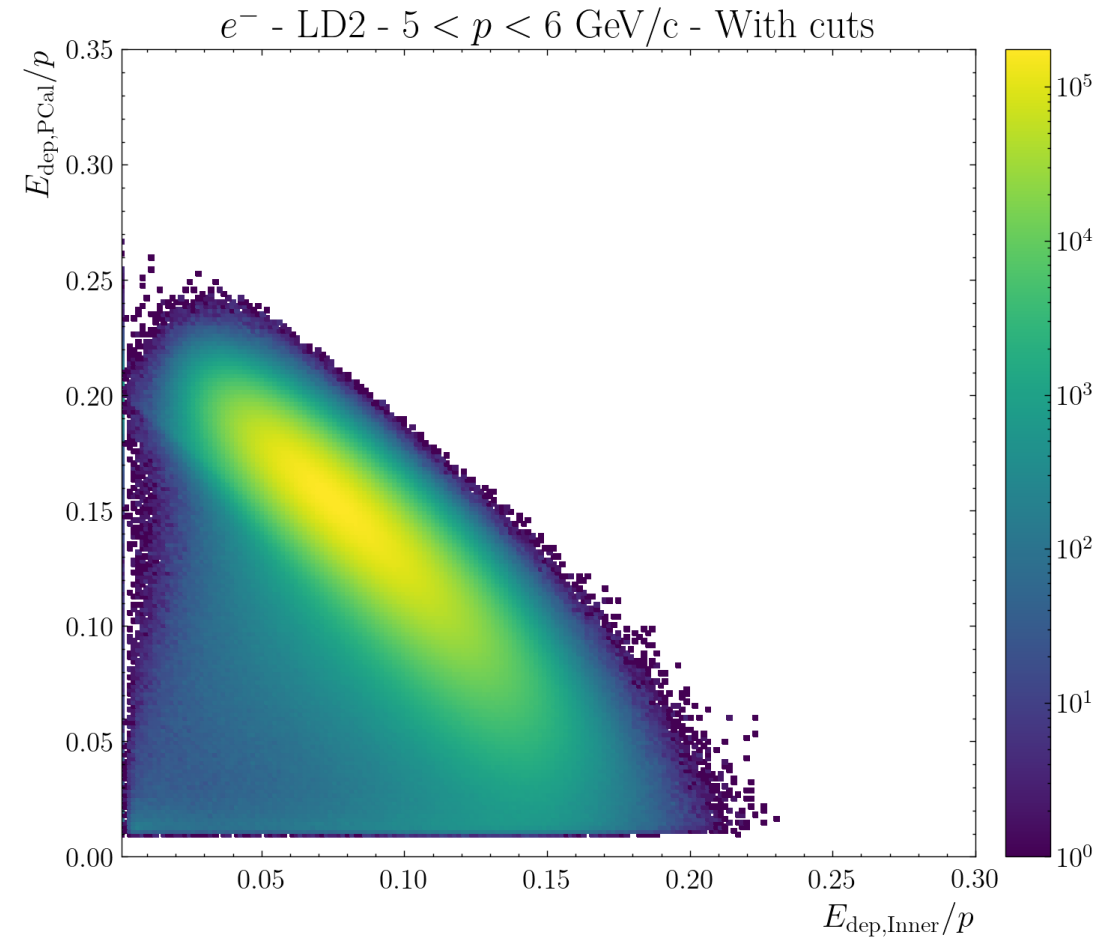
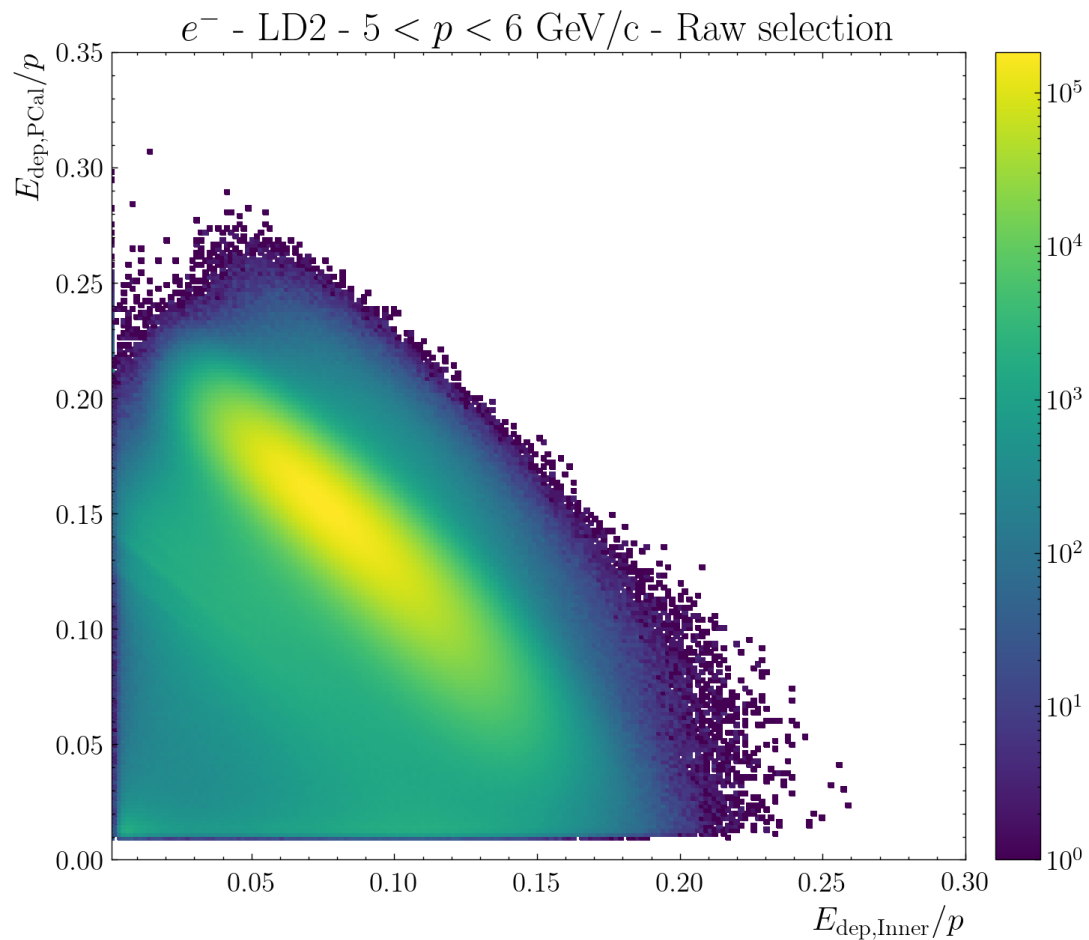


e^- - LD2 - After fiducial



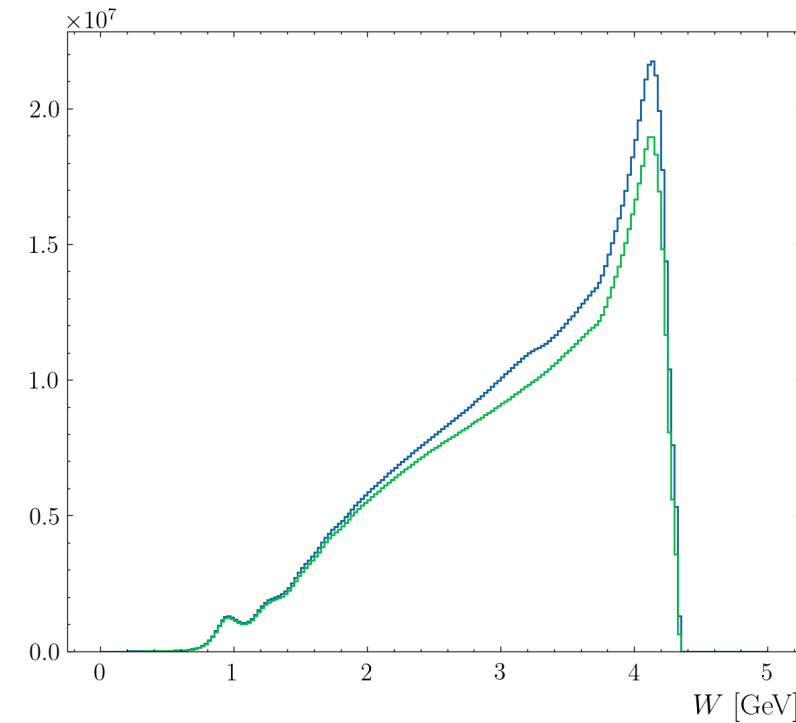
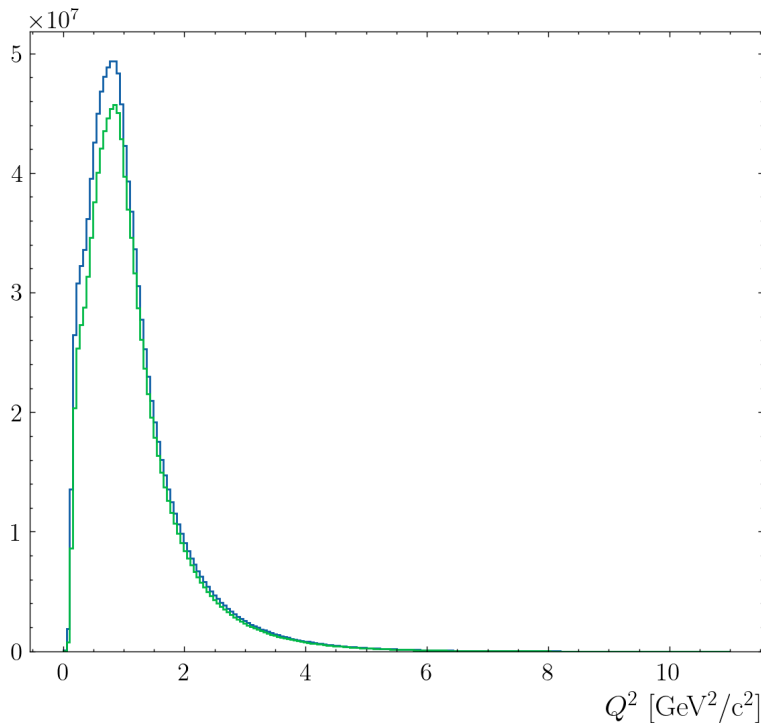
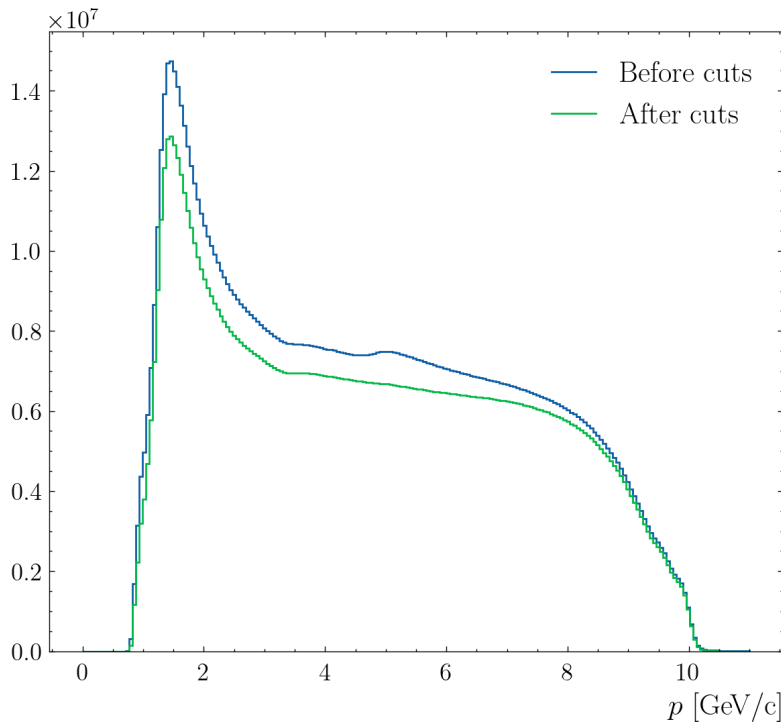
Electron identification: Triangular cut

- RG-A/B developed a cut on the sampling fraction of the inner calorimeter and the PCAL: Goal is to reduce the π^- contamination in the electron sample
- For RG-D, we found that the other cuts (SF and fiducials) are enough



Effect of the refinement cuts

- The refinement and fiducial cuts on the electron, don't reduce the phase space, only improving the electron sample
- The electron cuts are finalized, and an RG-D Common Analysis note have been started



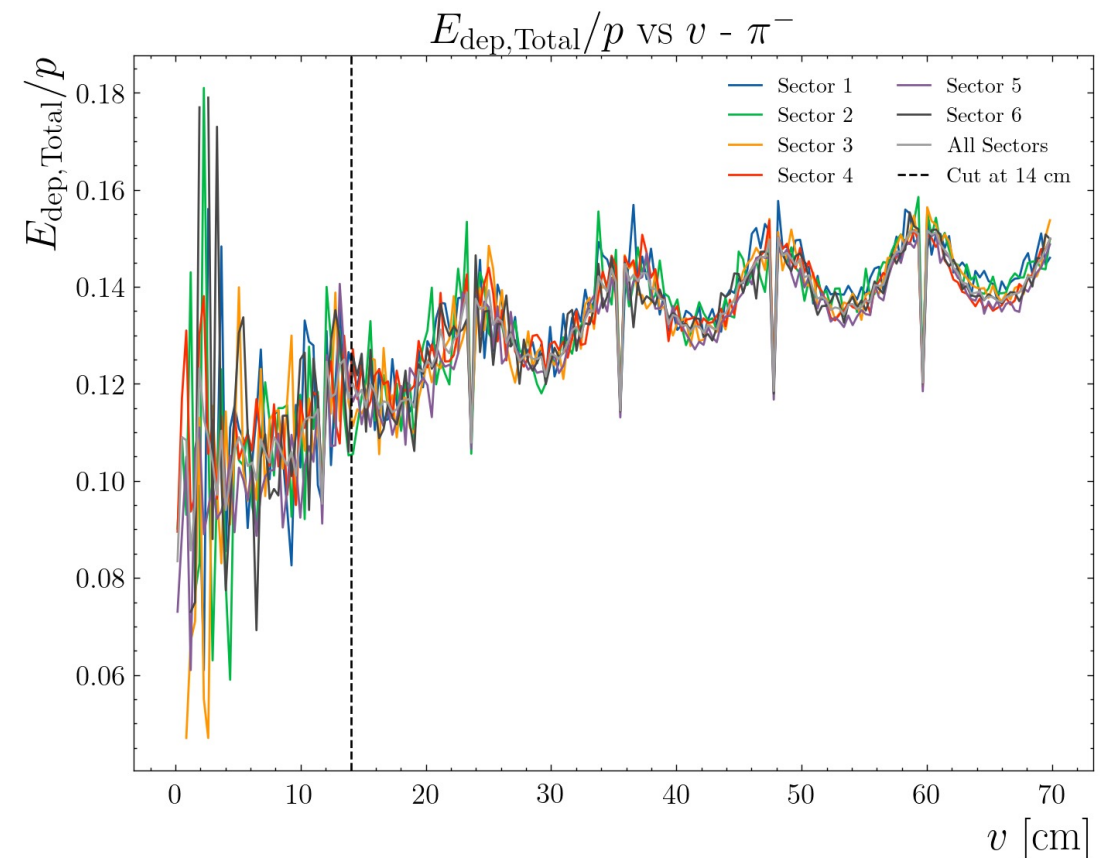
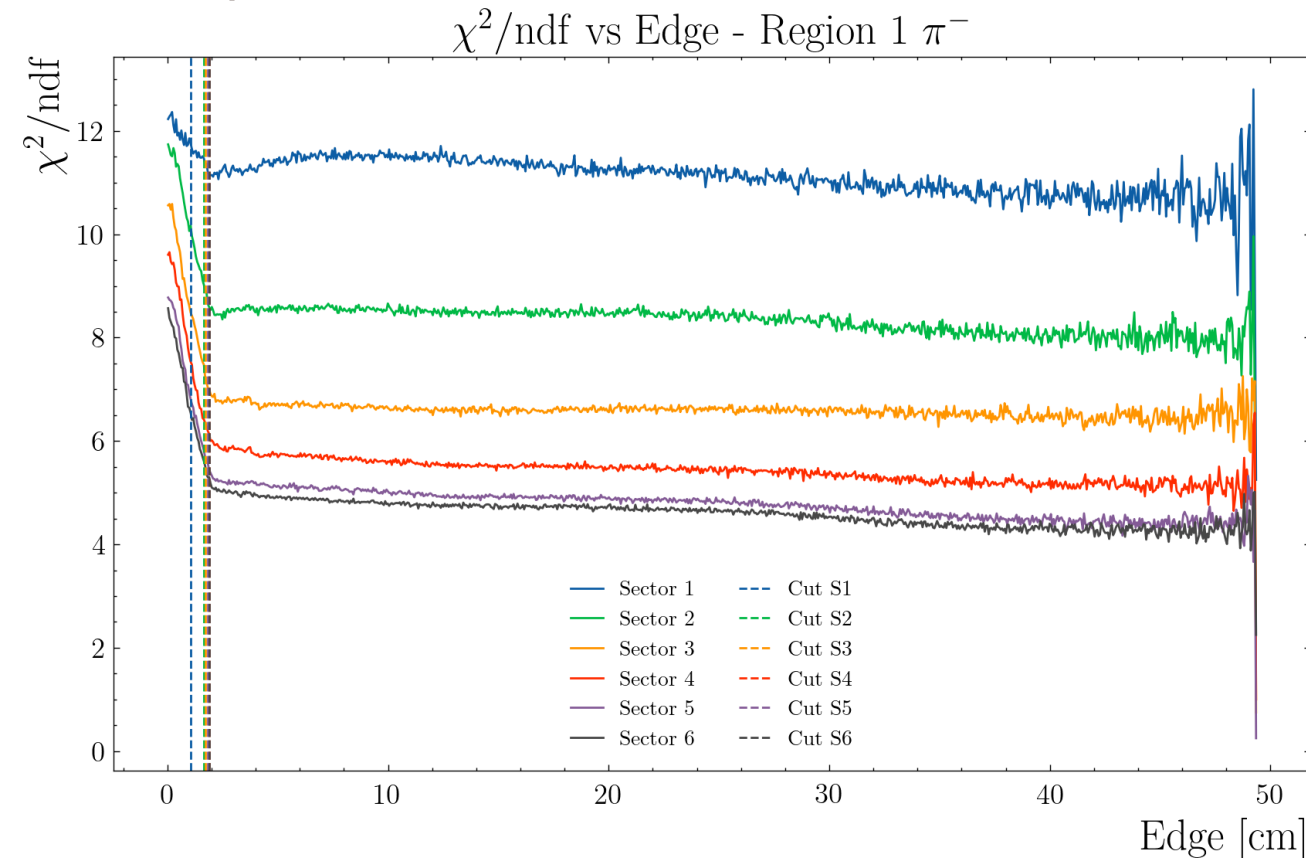
RG-D COMMON ANALYSIS NOTE

1	Particle identification and fiducial cuts	1
1.1	Electron identification	1
1.2	Electron fiducial cuts	6

1.2.1	Calorimeters fiducial cuts . . .	6
1.2.2	DC fiducial cuts	9

Hadron identification cuts

- Improve the hadron identification for pions (both for CT and TMD studies)
 - Calorimeter fiducial cuts: $v, w - 14$ cm
 - DC fiducial cuts: edge - 1.67, 3.64, 8.86 cm (target and region dependent)
 - Study hadron contamination: Presentation by Suman Shrestha



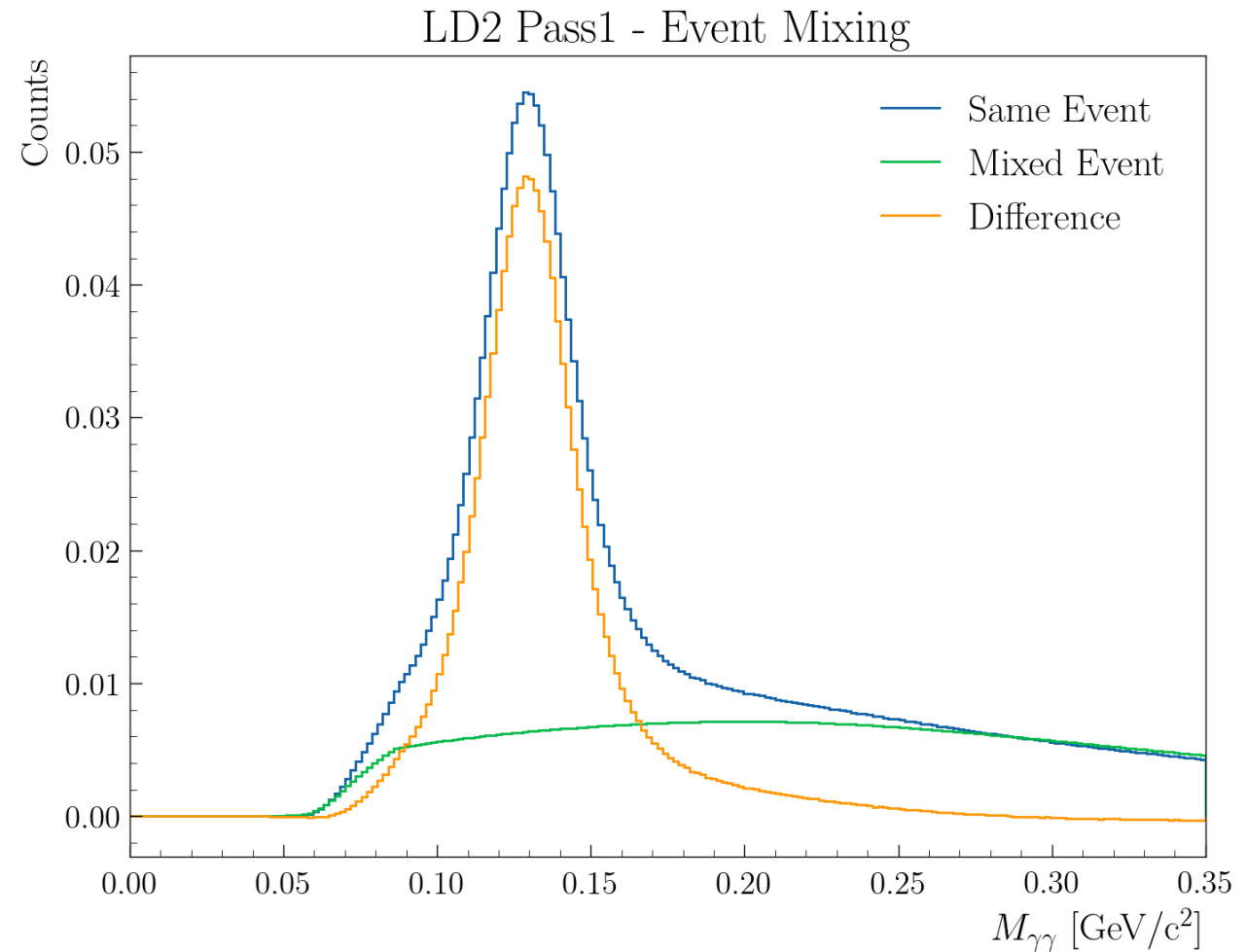
RG-D nTMDs Studies: π^0 analysis

- First look at the π^0 channel to extract multiplicity ratios and azimuthal asymmetries:

In-progress efforts to employ the event mixing technique to subtract the background underneath the two-photon mass peak

Electron cuts: pid = 11, status < 0, $2000 \leq |\text{status}| < 4000$, $|\text{chi2pid}| < 5$ and Vz cut

Photon cuts: pid = 22 and $E > 0.2$ GeV
+ AI develop by Gregory Matousek



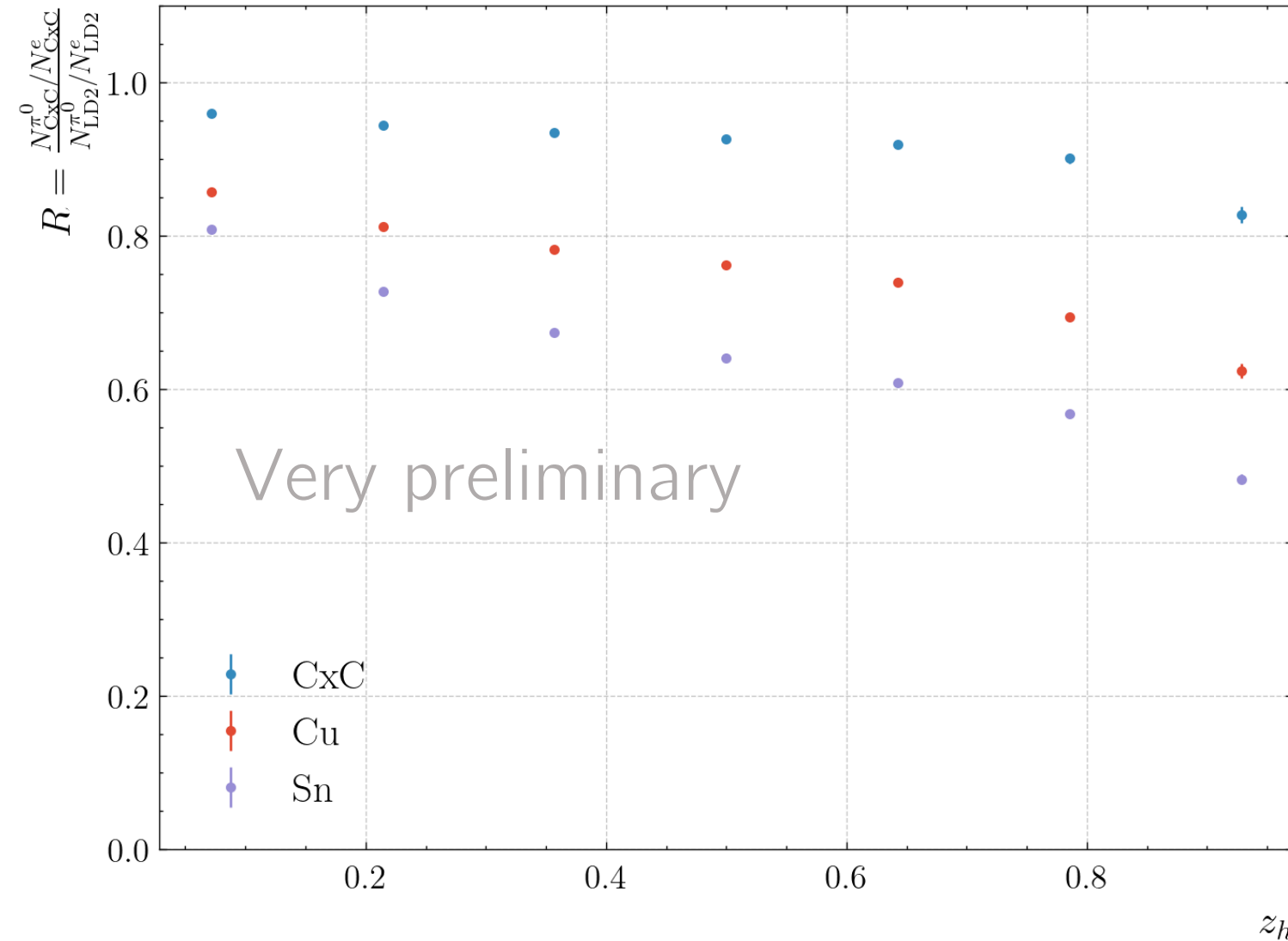
RG-D nTMDs Studies: π^0 analysis

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- 1D multiplicity ratio: z_h

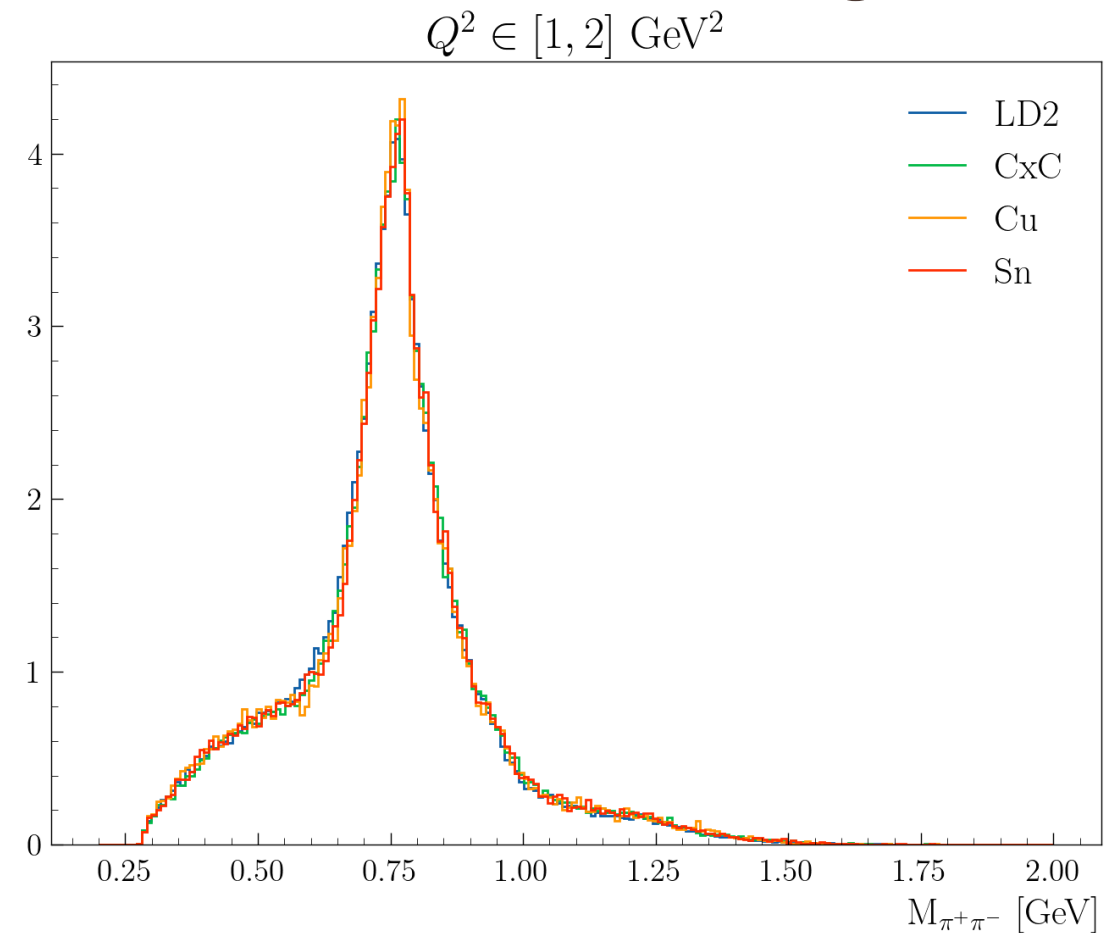
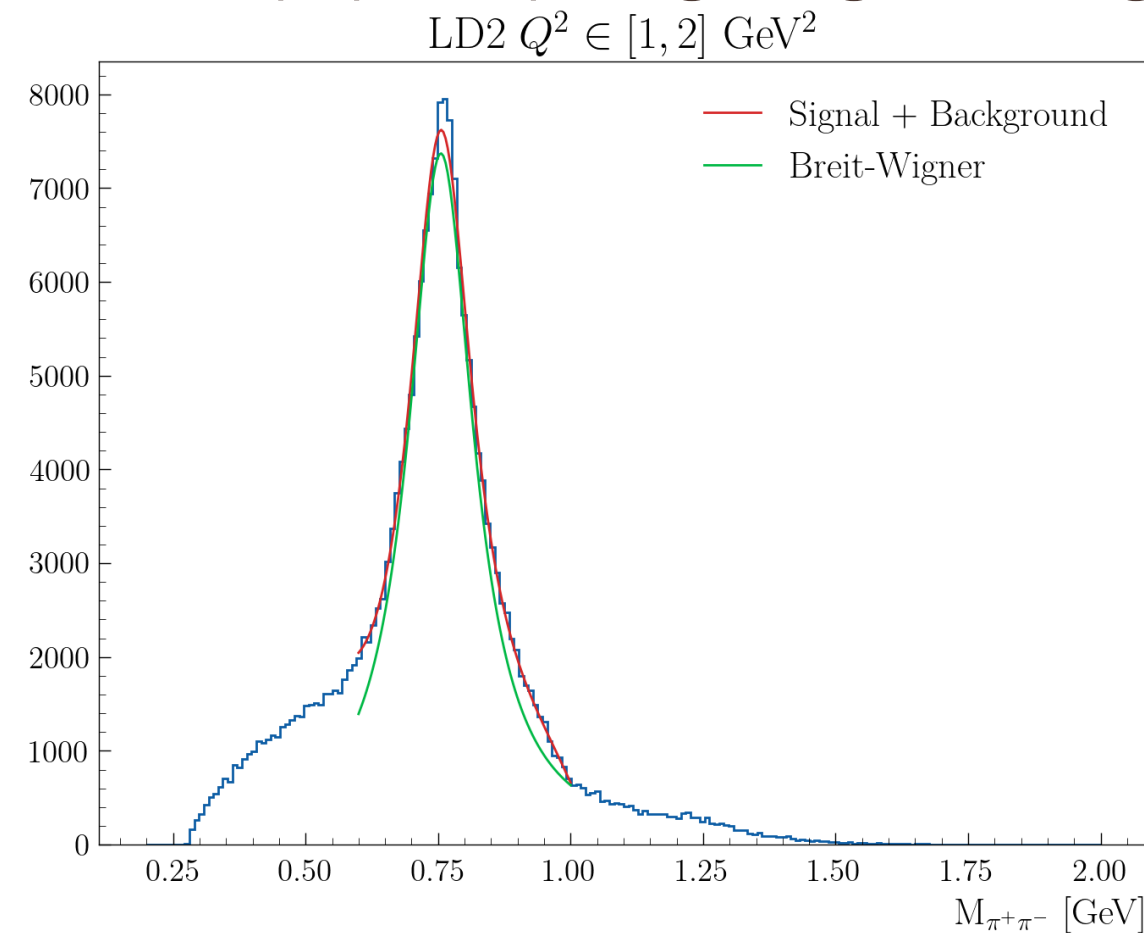
$$R = \frac{N_A^{\pi^0}(z_h)}{N_A^e} \frac{N_{LD2}^e}{N_{LD2}^{\pi^0}(z_h)}$$

- Statistical error only
 - SIDIS cut: $Q^2 > 1 \text{ GeV}^2$, $W > 2 \text{ GeV}$
- Goal: multidimensional ratios



CT Study: Very Preliminary Nuclear Transparency

- Fit oppositely charged pions invariant mass with a Breit-Wigner and 3rd-order polynomial function (*the latter will be replaced with the simulated background shape*)
- Extract ρ^0 yield by integrating the background-subtracted BW within a 3σ range



Ongoing analysis by M. Maynes

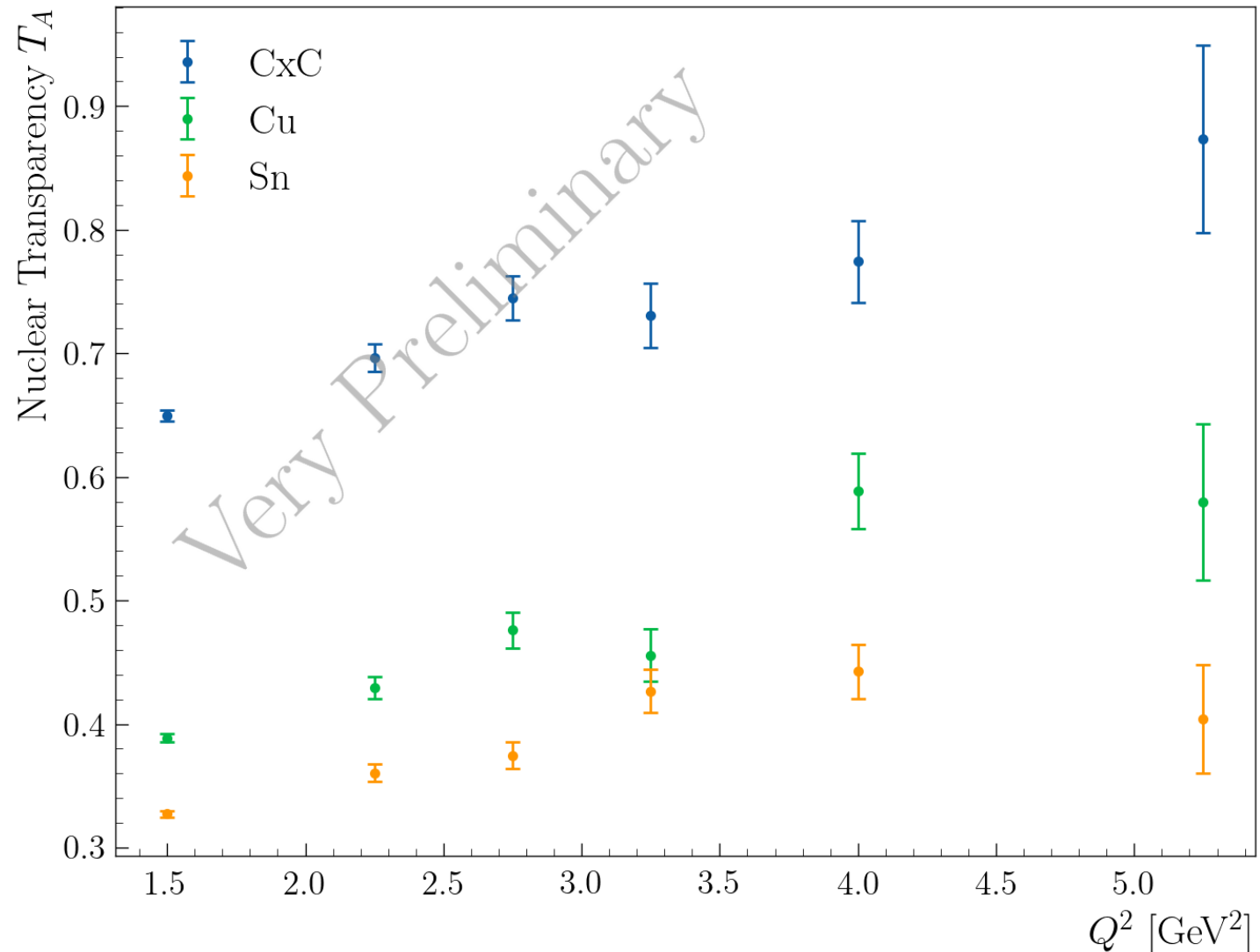
CT Study: Very Preliminary Nuclear Transparency

- Extract (raw) nuclear transparency with currently available Pass1 data (around 10% of total)

$$T_A = \frac{N_{\rho^0}^A}{N_{\rho^0}^{LD2}} \frac{r_{LD2} \rho_{LD2}}{r_A \rho_A}$$

- r_{LD2} = 5 cm: LD2 thickness
- r_{CxC} = 0.4 cm: CxC thickness
- r_{Cu} = 93 μ m: Copper thickness
- r_{Sn} = 171 μ m: Tin thickness
- ρ_{LD2} = 0.164 g/cm³: LD2 density
- ρ_C = 2.2g/cm³: Carbon density
- ρ_{Cu} = 8.96 g/cm³: Copper density
- ρ_{Sn} = 7.31g/cm³: Tin density

Ongoing analysis by M. Maynes



RG-D nTMDs Studies: π^+ analysis

- Motivation

- **SIDIS**: Electron scattering off a target nucleus producing hadrons through virtual photon exchange. Nuclear effects influence SIDIS behavior in nuclear targets
- **TMDs**: provide a 3D momentum-space description of quarks sensitive to transverse motion. This framework allows to separate initial and final state effects
- **Goal**: compare nuclear targets to deuterium reference to study medium induced modifications

$$R_A^h(Q^2, x_B, \phi_h, p_T^2, z) = \frac{\left(\frac{N_A^h(Q^2, x_B, p_T^2, z)}{N_A^e(Q^2, x_B)} \right)}{\left(\frac{N_D^h(Q^2, x_B, p_T^2, z)}{N_D^e(Q^2, x_B)} \right)}$$

- Analysis Tools

- Adapted SIDIS simulation for RG-D set up with nuclear target configuration
- Unfolding corrections using RooUnfold through custom response matrices

$$\Delta \langle p_T^2 \rangle(x_B, Q^2, \phi_h, z) = \langle p_T^2 \rangle_A(x_B, Q^2, z) - \langle p_T^2 \rangle_D(x_B, Q^2, z)$$

- Main observables

- Hadron multiplicity ratio
- Transverse momentum broadening $\Delta \langle p_T^2 \rangle$
- Azimuthal modulations over hadronic angle ϕ_h
- Study in multi-dimensional bins (Q^2, x_B, p_T^2, z) covering different kinematic ranges

$$\frac{\langle \cos 2\phi_h \rangle_A}{\langle \cos 2\phi_h \rangle_D} = \frac{\langle \cos 2\phi_h \rangle(x_B, Q^2, p_t^2, z)_A}{\langle \cos 2\phi_h \rangle(x_B, Q^2, p_t^2, z)_D}$$

- Ongoing analysis by D. Matamoros

Summary

- RG-D has passed the Pass1 review, and the cooking continue
 - Cooking will take few more months
- Optimizing analysis tools for CT and nTMDs studies to
 - Improve Particle Identification and implement fiducial cuts for all final-state particles
 - Study the hadron/electron contamination in pion sample
 - Correct for contamination with the CuSn target
 - Deploy the ρ^0 event generator for the two-pion invariant mass background subtraction and apply necessary corrections for the extraction of the preliminary CT results
 - Finalize the π^0 event mixing background-subtraction and extract its preliminary asymmetry results
 - Obtain the preliminary asymmetry results for charged pions nTMDs studies

The background is a dark blue gradient. It features several faint, stylized atomic models with circular orbits and a central nucleus. A prominent starburst of light rays emanates from the center-left. Two clusters of red and blue spheres, representing nucleons, are positioned on the left and top right. A particle track with a vertex is visible in the bottom right.

THANKS